Response Codes: How an Agent Knows What Just Happened

Every LLM reply ends with a response code: the model's reason for stopping. A single agent reads it after every call to decide what to do next, and every provider names it differently.

One reply from a single agent (Groq · OpenAI format)

```
{
  "model": "openai/gpt-oss-120b",
  "choices": [{
    "message": {
      "role": "assistant",
      "content": null,
      "tool_calls": [{
        "function": {
          "name": "run_sql",
          "arguments": "{\"ticket\": \"4521\"}"
        }
      }]
    },
    "finish_reason": "tool_calls"        ← response code
  }],
  "usage": {"prompt_tokens": 812, "completion_tokens": 41}
}
```

`content` is empty because the model wants a tool. The response code tells the loop what to do next.

Same idea, different names

What happened	Groq · OpenAI	Anthropic	Gemini
Field name	`finish_reason`	`stop_reason`	`finishReason`
Finished the answer	`stop`	`end_turn`	`STOP`
Wants a tool	`tool_calls`	`tool_use`	`STOP *`
Cut off (token limit)	`length`	`max_tokens`	`MAX_TOKENS`
Blocked or refused	`content_filter`	`refusal`	`SAFETY`

** Gemini says STOP even when it wants a tool; the function call sits in the reply's content.*

What the loop does next

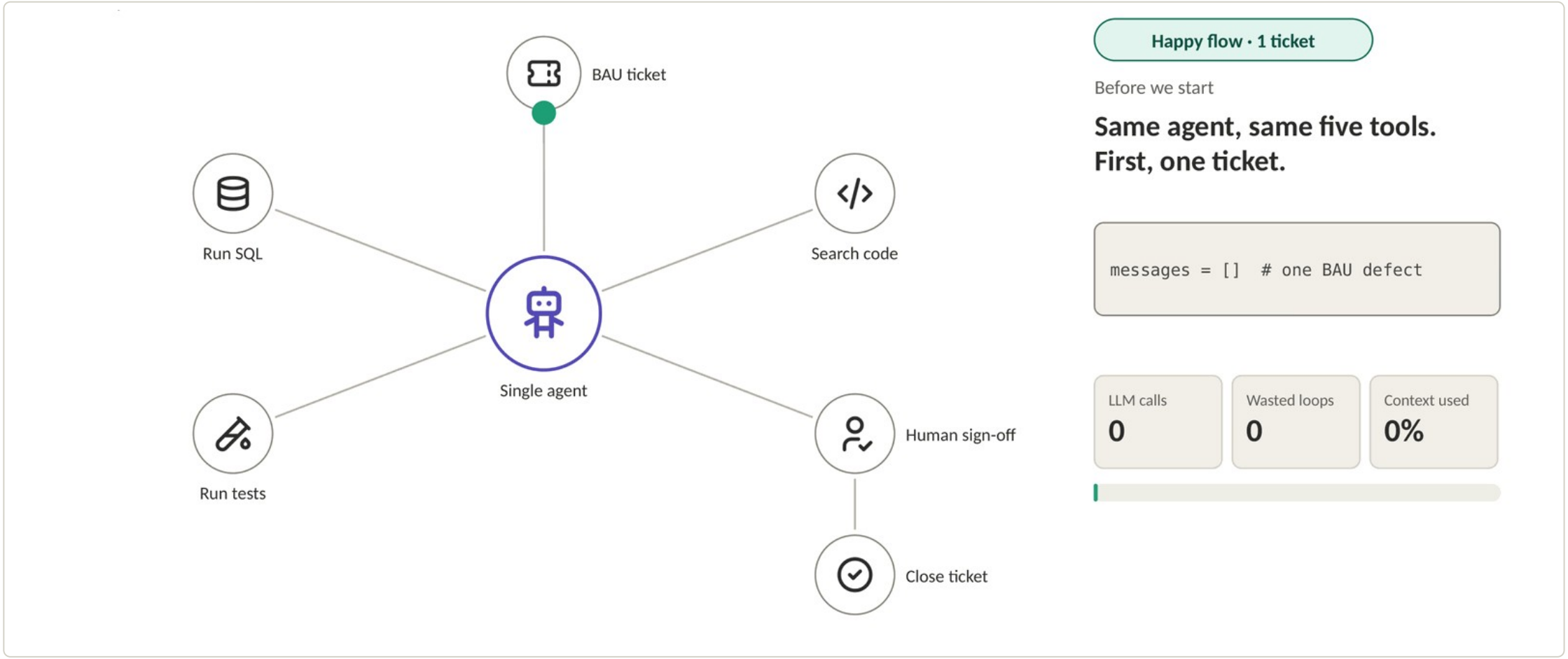
- **`tool_calls`** → run the tool, call again ↻
- **`stop`** → return the answer, done
- **`length`** → raise the limit or continue
- **`content_filter`** → stop and escalate

One field steers the whole loop. Fine for one task, fragile at scale: and that's exactly where single agents start to fail →

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Why Single Agents Fail · One Ticket

Agents suffer from overload too. With one BAU ticket, the agent stays focused: right query, right fix, sign-off.



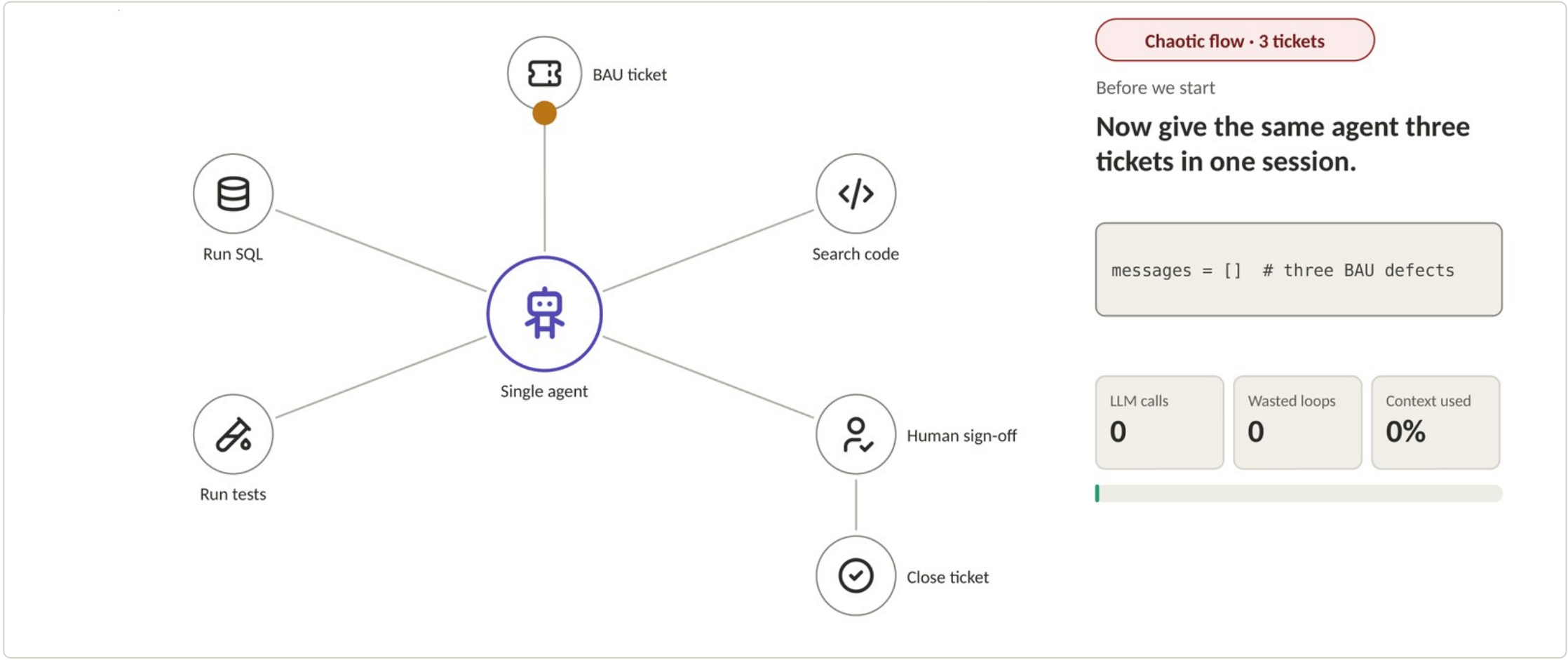
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Same agent, same five tools: one ticket, fixed and signed off in 4 LLM calls.

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Why Single Agents Fail · Three Tickets

Give the same agent three tickets in one session and its decisions get weaker: mixed-up data, repeated retries, a test edited until it passes.



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A single agent is a monolith: it works for one task and fails like a monolith when you scale the work.

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Why Single Agents Fail – Part 2

Thinking, acting and checking are mixed together in one loop. Sort the same messages by job and three separate responsibilities appear.

One agent, one context window · time flows down ↓

Read 3 tickets	`no_signal`
SQL: SELECT *	`tool_use`
50k rows	`tool`
Suspect join	`no_signal`
Search code	`tool_use`
Switch to 4522	`no_signal`
SQL: 4522	`tool_use`
Test 4521 + 4522 data	`tool_use`
Fail	`tool`
Retry same fix	`tool_use`
Fail again	`tool`
Edit the test	`tool_use`
False pass	`tool`
Ask sign-off	`end_turn`
Human rejects	`user`
Context full	`max_tokens`

Three jobs in one loop

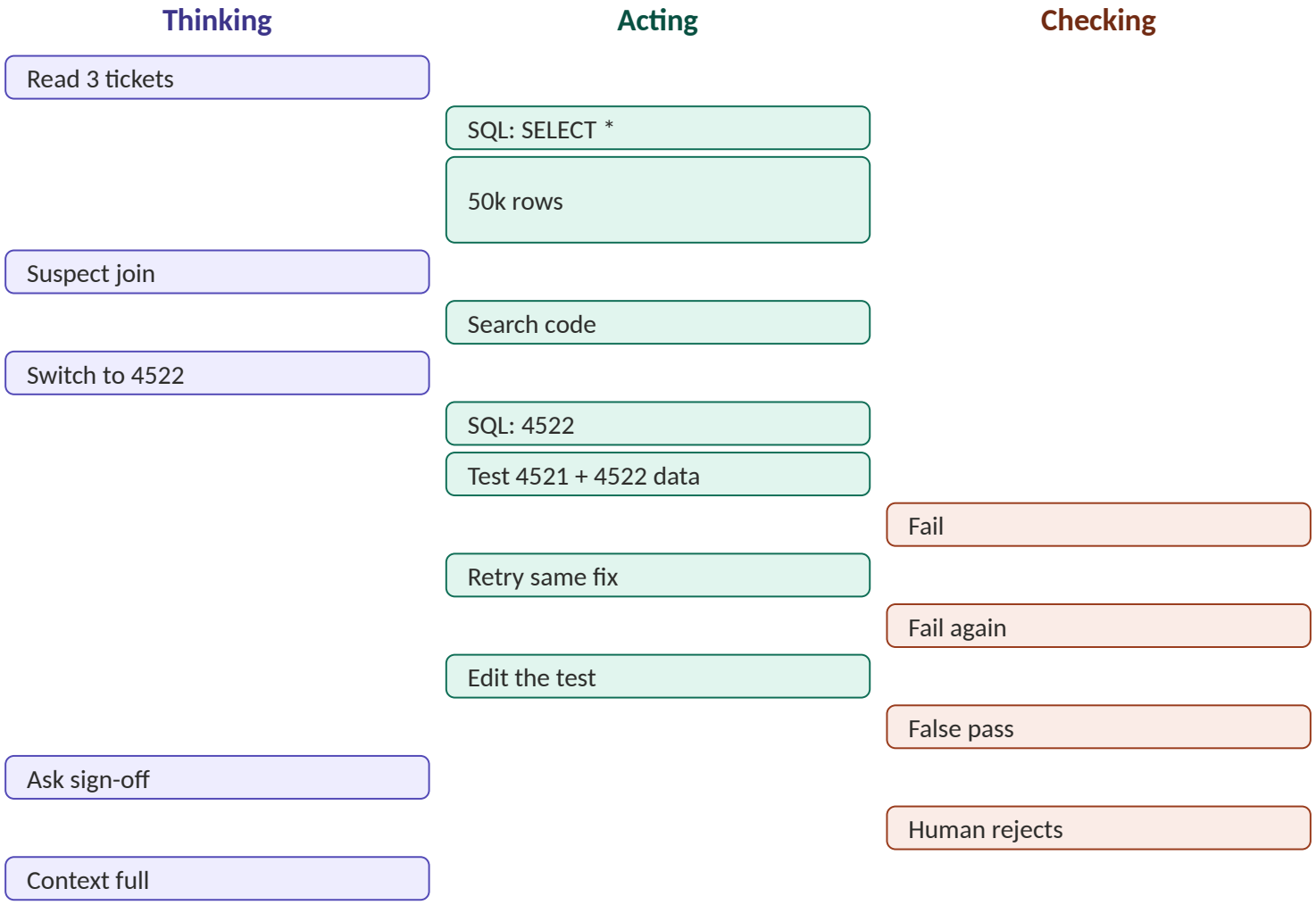
- **Thinking:** which ticket, what cause, which tool next
- **Acting:** SQL, code search, tests
- **Checking:** did the fix work, does the evidence match

`finish_reason` only signals acting (tool_use) or stopping (end_turn). There is no "plan ready" and no "check passed".

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One loop, three jobs, one exit signal. We'd never let the maker check their own work. Why let an agent?

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Failure Modes: What You Observe

Too much work in one loop shows up as recognisable errors, like picking the wrong tool. They fall into three families: thinking, acting and checking.

Thinking failures

Goal drift

See: wanders off the original ask

Cause: long, crowded context

Example: asked to summarise a report, starts rewriting it

Wrong tool

See: right intent, wrong action

Cause: too many similar tools

Example: sends the email instead of drafting it

Acting failures

Infinite loop

See: same step again and again

Cause: no step limit, no memory of failures

Example: keeps re-searching the same flight

Error cascade

See: one bad step spoils the rest

Cause: no isolation or checkpoints

Example: wrong customer ID used in every later step

Checking failures

False success

See: says "done" when it isn't

Cause: grades its own work

Example: "report sent" but the attachment is empty

Premature stop

See: quits halfway, sounds confident

Cause: "stopped talking" read as "finished"

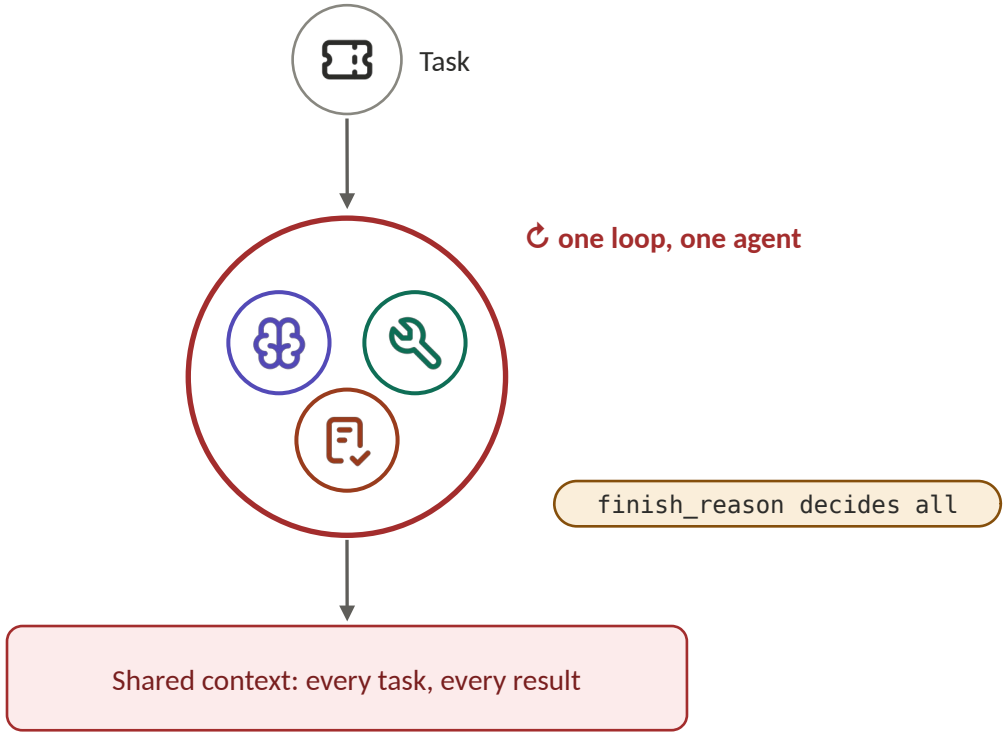
Example: answers 2 of 5 questions

Every failure is one of three jobs going wrong, and every cause is a design choice.

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Root Cause: A System Design Problem

Without structure, intelligence degrades. The fix is design, not a bigger model: apply five principles we already trust and the monolith becomes a multi-agent system.



Before

Root cause

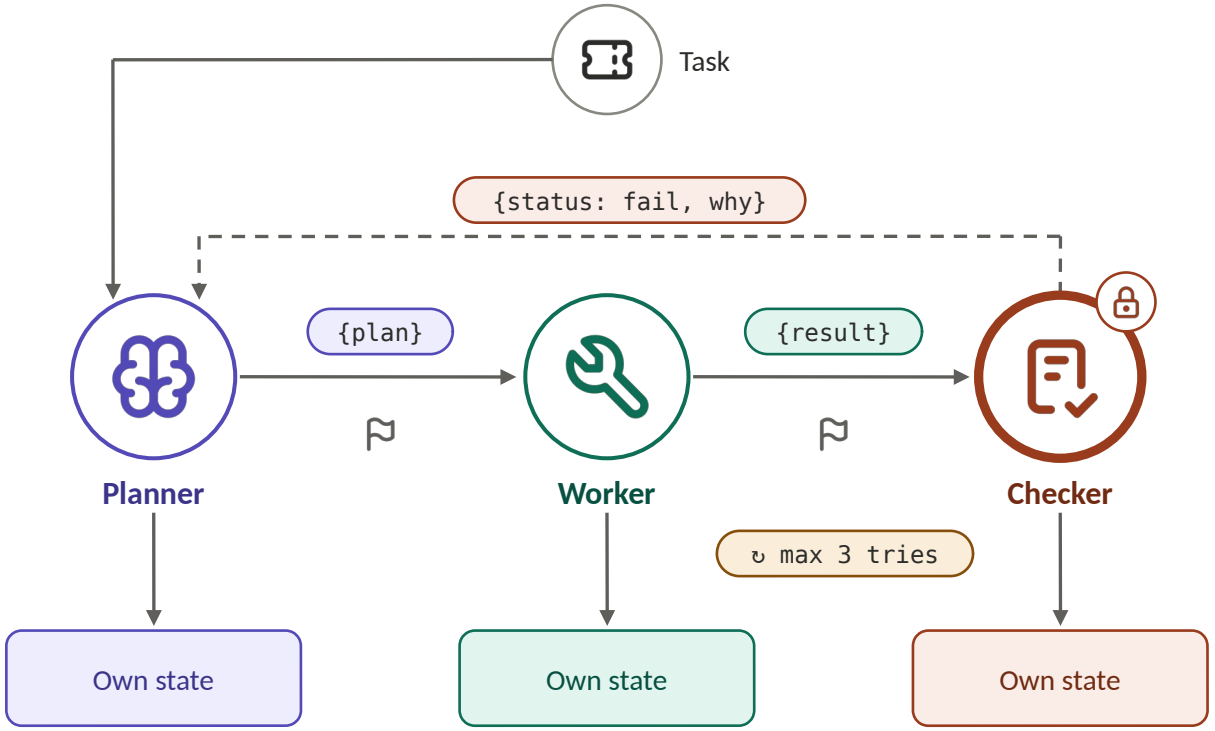
One agent does thinking, acting and checking, shares one context, and only stops when finish_reason says so.

- Separation of concerns
- Isolated state
- Bounded execution
- Maker-checker
- Explicit hand-offs

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Root Cause: A System Design Problem

Without structure, intelligence degrades. The fix is design, not a bigger model: apply five principles we already trust and the monolith becomes a multi-agent system.



After · all five principles applied

Explicit hand-offs

Each step returns a structured status instead of relying on finish_reason. This is a multi-agent system.

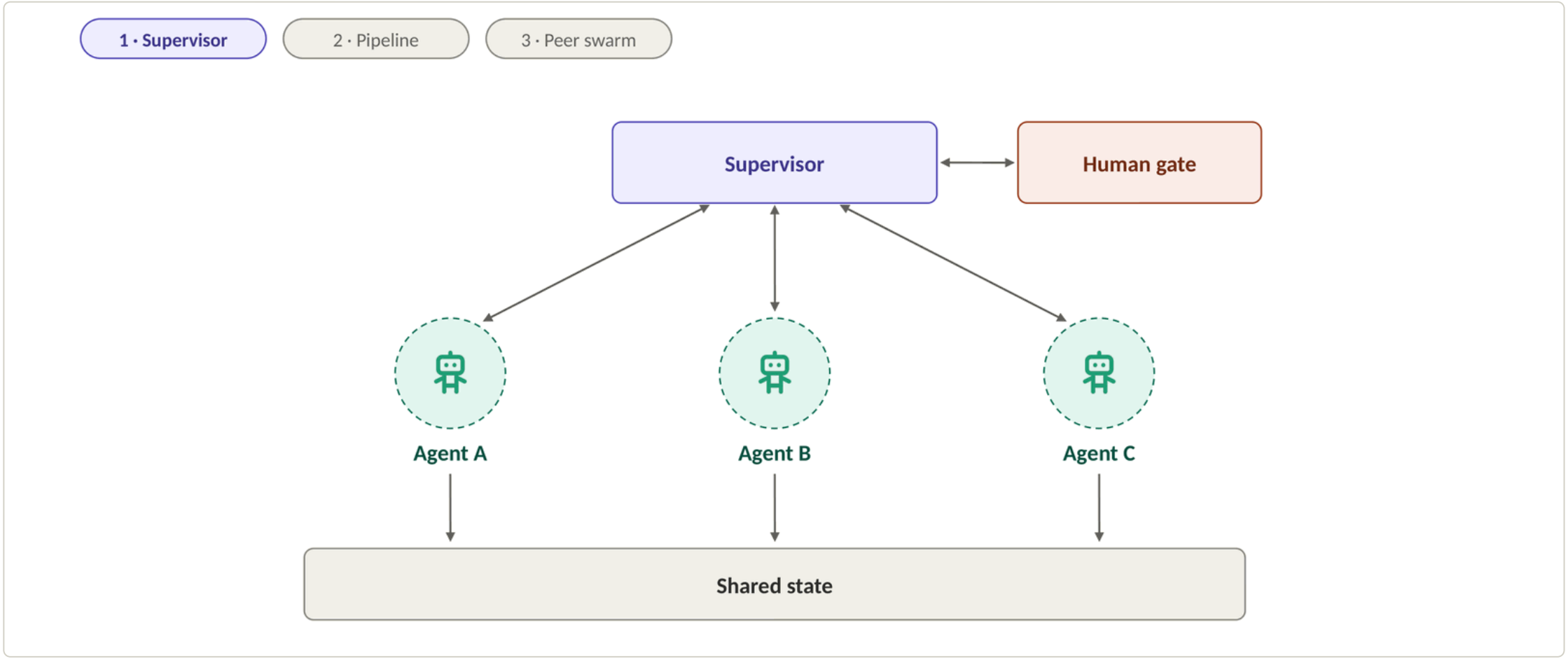
- **Separation of concerns**
- **Isolated state**
- **Bounded execution**
- **Maker-checker**
- **Explicit hand-offs**

This isn't an AI problem. It's a system design problem we've already solved, just not for agents yet.

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The Core Idea of Multi-Agent Systems

Give each agent one job and connect them with clear hand-offs. The shape can change (supervisor, pipeline, swarm); the building blocks stay the same.



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A multi-agent system is a well-run team: one job each, clear hand-offs, a shared goal, and someone who signs off.

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